

Iteris — Evaluation Report

Plain-language summary of the 2026-07-20 evaluation pass · DRL contour refinement for medical image segmentation · Taylor's University, PRJ63504 Capstone

Executive Summary

- We trained two AI agents (DuelingDDQN and TD3) to nudge the boundary a U-Net model draws around a heart structure (CAMUS ultrasound) or a brain tumour (BRISC MRI), one small step at a time.
- Across 16 test runs, the agents rarely beat a **strong** starting U-Net, but did make a real, measurable improvement over a **weaker** starting U-Net on the heart scans.
- The single most important number: how good an agent's final answer is correlates 0.98–1.00 with how good the U-Net's starting answer already was. In plain terms — the agent is mostly riding on the U-Net's coattails, not yet adding independent judgement of its own.
- No statistical significance testing (e.g. a paired Wilcoxon test) has been run on this pass — every number below is a point estimate from a single evaluation run, not a tested claim.

What Was Tested

Two datasets (CAMUS heart ultrasound with 3 structures — LV endocardium, LV epicardium, left atrium; and BRISC brain MRI tumour), two starting points (a strong **Attention U-Net** and a weaker, lighter **Lite U-Net**), and two agents (**DuelingDDQN**, which picks from a fixed menu of moves, and **TD3**, which picks any move on a continuous dial). That's 2 agents × 2 baselines × 4 structures = 16 test runs in total.

Results at a Glance

Dice measures overlap with the correct answer (1.000 = perfect match). HD95 measures boundary distance error in millimetres/pixels — lower is better. Δ is the change vs. that phase's own baseline.

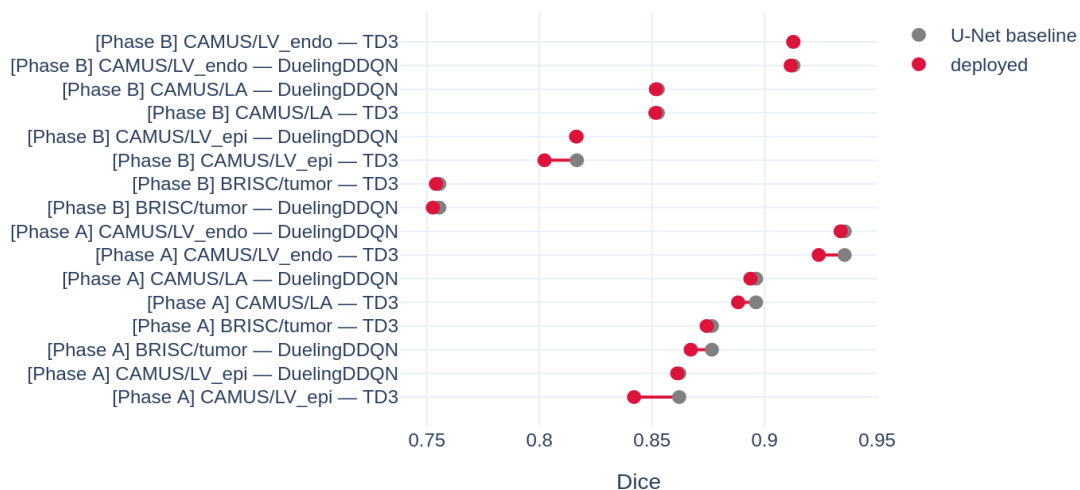
Class	Baseline Dice	DuelingDDQN Dice	Δ	TD3 Dice	Δ
Phase A — vs. Attention U-Net (strong baseline)					
BRISC / tumor	0.870	0.867	−0.2pp	0.874	+0.5pp
CAMUS / LV endo	0.936	0.934	−0.2pp	0.924	−1.2pp
CAMUS / LV epi	0.869	0.861	−0.8pp	0.842	−2.7pp
CAMUS / LA	0.893	0.894	+0.0pp	0.888	−0.5pp
Phase B — vs. Lite U-Net (weak baseline)					
BRISC / tumor	0.819	0.753	−6.6pp	0.754	−6.5pp
CAMUS / LV endo	0.910	0.912	+0.1pp	0.913	+0.2pp
CAMUS / LV epi	0.809	0.816	+0.7pp	0.802	−0.7pp
CAMUS / LA	0.821	0.852	+3.0pp	0.852	+3.0pp

Source: master_comparison.csv, full 2026-07-20 evaluation run. "Unknown"-phase baseline rows were resolved by dice rank (higher = Phase A / Attention U-Net, lower = Phase B / Lite U-Net), cross-checked against every DRL run's own recorded baseline dice — 16/16 exact matches.

What Went Well

- On the heart scans (CAMUS) against the **weak** starting point, both agents made real improvements — the left atrium boundary gained about **+3.0 percentage points** of Dice for both agents, the largest clean win in the whole evaluation.
- The discrete (menu-of-moves) and continuous (dial) agent designs land in almost the same place on every run — the head-to-head scatter plot below sits right on the diagonal. That means the choice of action space isn't what's holding results back.
- The safety-net design (the deployed agent is never allowed to score worse than the U-Net's own starting answer) worked as intended across the runs — it's exactly why the reported “deployed” numbers never show a catastrophic collapse, only modest gains or modest losses.

Before (baseline) -> after (deployed) Dice, every run (grouped by phase)

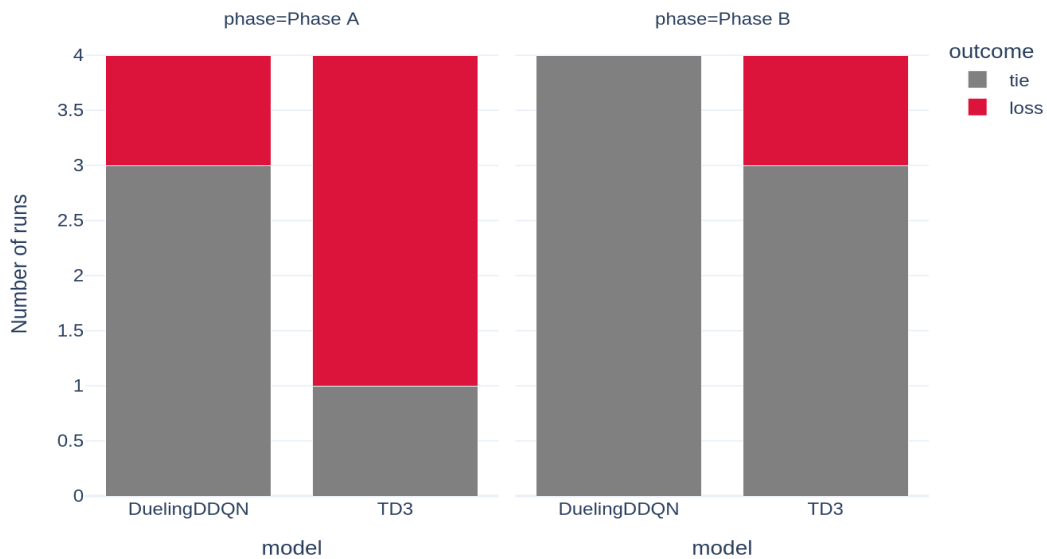


Baseline → deployed Dice for every one of the 16 runs, grouped by phase.

What Went Wrong

- On the brain scans (BRISC) against the **weak** starting point, both agents did clearly worse — Dice dropped by about **6.5–6.6 percentage points**, and the boundary-distance error (HD95) roughly doubled. This is the one unambiguous regression in the study.
- Against the **strong** starting point (Attention U-Net) on any structure, neither agent reliably beats it — results are flat to slightly negative everywhere except a marginal TD3 win on BRISC.
- Some runs show “drift”: the agent's best moment during training is better than where it ends up by the end of the episode, meaning it can overshoot a good answer rather than stopping there.

Win / tie / loss per agent, by phase (tie band = ± 0.005)



Win/tie/loss vs. the paired baseline, by phase (tie band = ± 0.005 Dice).

Why It Happened

Three separate limits get conflated whenever a refinement agent “fails to beat the baseline”. All three show up consistently in this evaluation:

1. The output format itself has a built-in cost

The agent's output is a smoothed curve fitted to the largest connected region of the mask — it can never reproduce the U-Net's raw mask exactly, even if it “does nothing” useful. That's a small penalty paid just for changing representations, before the agent has learned anything at all.

2. A good starting point leaves little room to improve

Against the strong Attention U-Net, there just isn't much error left to correct — you can't win a lot from an already-good answer. Against the weaker Lite U-Net, there's more slack, which is exactly where the agents show their one clear win (CAMUS left atrium).

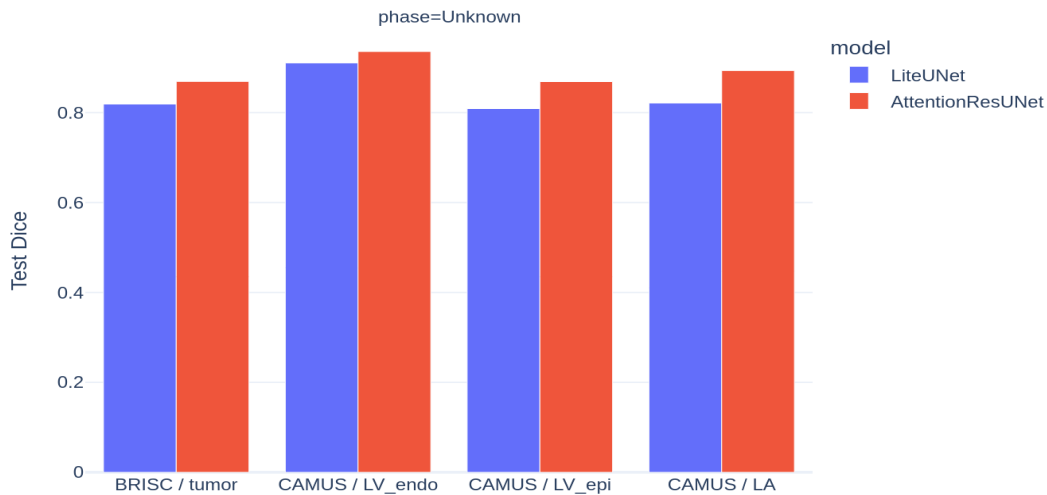
3. No learned “stop now” signal

TD3 in particular has no trained signal for “this is as good as it gets, stop moving.” A policy can keep nudging the boundary past its best point before the episode ends, which is the drift behaviour noted above.

The decisive evidence

Across all 16 runs, how good the **deployed** (final) answer is correlates **0.98 (Phase A)** and **1.00 (Phase B)** with how good the **starting** U-Net answer already was. In plain terms: almost all of the variation in the final score is explained by where the agent started, not by anything it learned along the way.

U-Net baselines by dataset / class, split by phase



The two U-Net starting points compared directly — the gap the agents have to work with.

What the Model Learned vs. Didn't Learn

Learned:

- Small, safe corrective nudges when the starting point already has some slack (weak baseline, easier structures).
- The discrete-menu and continuous-dial move choices are roughly interchangeable — both reach the same place.

Not learned (yet):

- How to meaningfully correct a strong baseline's mistakes.
- Reliable behaviour on the hardest case in this study — BRISC tumour against the weak baseline — which is the one clear regression.
- When to stop moving — there is no learned stopping signal, especially for TD3.

Does This Evaluation Compare Phase A vs. Phase B Directly?

No — not as a direct, tested comparison. The evaluation output only ever compares each phase's DRL agents against *that phase's own paired baseline* (Phase A agents vs. Attention U-Net; Phase B agents vs. Lite U-Net). There is no row or statistical test in the source data that compares Phase A's numbers to Phase B's numbers head-to-head, and no significance test (Wilcoxon or otherwise) was run between phases.

That said, because both phases' numbers are known, a side-by-side reading is possible and is what this report (and the Results table on the Research page) does — e.g. “DuelingDDQN scores 0.860 mean Dice in Phase B vs. 0.896 in Phase A on CAMUS.” That's a descriptive comparison, not a tested one.